

Aidan Law

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U.S. Citizen | Eligible for Security Clearance

EDUCATION

University of Maryland

Bachelor of Science in Mechanical Engineering, Minor in General Business

- **GPA:** 3.63/4.00 | **Dean's List** (Fall 2024 – Spring 2026)

College Park, MD

Sep. 2024 – May 2028

EXPERIENCE

Manufacturing Engineering Intern

May 2026 – Present

MW Components

Fairfield, NJ

- Developed C# application using SolidWorks API to automate hardware 3D CAD modeling, 2D drawing generation and review, and quality checklists, **reducing lead times by 99% (from 45 min to 30 sec)** across **1,000+ parts**.
- Programmed C# math routines to automatically recalculate ASME/ISO thread geometry based on surface coating/plating thickness, eliminating manual calculations for 5-axis CNC manufacturing drawings.
- Authored ISO 9001/AS9100D Calibration SOP and developed Python inventory optimization algorithms, **slashing external calibration costs by 75%** and **increasing shop-floor tool availability by 25%**.
- Translated complex customer blueprints for Tier 1 aerospace and defense prime contractors into floor-ready manufacturing drawings compliant with strict Geometric Dimensioning and Tolerancing (GD&T) standards.

Data Engineering Intern

Jun. 2025 – Aug. 2025

Publicis Health

Morristown, NJ

- Re-engineered a manual monthly reporting workflow into an automated data pipeline using Python, SQL, and VBA, **reducing process cycle time by 98% (from several hours to 5 min)** and eliminating recurring manual errors.
- Applied process optimization and root-cause analysis to client campaign performance data, writing Python scripts to automate targeted outreach to 2,500+ stakeholders and improve process efficiency at scale.
- Performed statistical analysis and data-driven decision support across 100,000+ data points, engineering client dashboards in Power BI and Excel to translate raw datasets into actionable process and pricing insights for cross-functional teams.

PROJECTS

Over-Terrain Autonomous Vehicle | C++, Arduino, Fusion 360, SolidWorks

- Awarded **1st place with a 120/100 score** for mechanical design, system integration, assembly, and prototyping of an autonomous mechatronic vehicle that completed course objectives under budget and within top-3 fastest run times.
- Programmed C++ Finite State Machine (FSM) embedded control logic on Arduino to automate machine sequencing, integrate HX711 strain gauge sensor data acquisition, and classify material density via force switch for real-time control.
- Engineered a low-friction laser-cut acrylic chassis and friction-ramp loading mechanism, wiring and troubleshooting electro-mechanical subsystems (motors, sensors, actuators) and conducting free-body torque and velocity analyses for component selection and 12 V motor sizing.

Drone Propeller Study | MATLAB, SolidWorks, FEA, Aerodynamics

- Developed a custom Blade Element Method (BEM) aerodynamic solver in MATLAB, modeling tip-loss tapering and induced power to evaluate 4 parametric blade geometries for a 1.25 lb drone operating at 8,500 RPM.
- Modeled parametric 3D drone propeller blades in SolidWorks based on MATLAB numerical output; conducted structural FEA under combined centrifugal and aerodynamic loads to optimize hub root stress concentrations.
- Executed a parametric trade study comparing Bullnose and Elliptical propeller profiles against commercial benchmarks, achieving an estimated 52% increase in battery and hover efficiency and a peak thrust of 1.9 lb.

Drill Press CAD Assembly | SolidWorks, FEA, Motion Analysis, Mechanical Design

- Modeled a 100+ component drill press assembly in SolidWorks across 5 unique subassemblies.
- Designed a custom rack-and-pinion handwheel drive and crank-driven table lift system with dynamic mechanical mates to simulate precise plunge motion, stroke travel, and subsystem interactions.
- Executed static FEA on the worktable under a 500 N downward drilling force, verifying zero material yield and minimal structural deflection under peak operational loads to ensure a high working safety factor.

Custom FEA Solver | C++, Python, CAD/CAE Integration

- Built a 3D Finite Element Analysis (FEA) solver in C++ and Python to model structural stress and deflection, achieving sub-second solution convergence across a 20x coarser mesh.
- Accelerated CAD simulation setup by 99.5% (from 120 sec to 0.6 sec) by formulating a C++ quadratic 10-node tetrahedral (TET10) engine via nanobind that eliminates artificial shear locking in bending.
- Engineered an interactive 3D desktop interface using native raycasting observers for real-time CAD surface selection.

TECHNICAL SKILLS

CAD & Design: SolidWorks (incl. API), Fusion 360, Autodesk Inventor & Revit, 3D Modeling, 2D Drawings & Review

Manufacturing: GD&T, DFM, CNC Machining, 3D Printing, Laser Cutting, ISO 9001, AS9100D, Lean Six Sigma Certified

Systems & Hardware: System Integration & Assembly, Mechanical Hardware Design, Embedded Systems, Arduino

Programming & Analytics: Python, C++, C#, Java, MATLAB, SQL, VBA, Power BI, Excel, PowerPoint, Word